# Normalisation

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# Normalization of Relations

**Normalization:**

- The process of decomposing unsatisfactory "bad" relations by breaking up their attributes into smaller relations

**Desired Normal form:**

- Condition (using keys and FDs of a relation) to certify whether a relation is in a particular normal form

# Second Normal Form (2NF)

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# Second Normal Form (2NF)

- A relation is in 2NF if it is in 1NF and if it does not contain partial functional dependencies
  - If a relation has a single-column primary key, then there is no possibility of partial functional dependencies. Such relation is automatically in 2NF and it does not have to be normalized to 2NF
  - If a relation with a composite primary key has partial dependencies, then it is not in 2NF, and it has to be normalized into 2NF

# Normalise Relation (AD CAMPAIGN MIX) into 2NF (1)

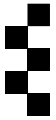
| A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|
| **AdCampaignID** | **AdCampaign Name** | **StartDate** | **Duration** | **Campaign MgrID** | **Campaign MgrName** | **ModusID** | **Media** | **Range** | **Budget Pctg** |
| 111 | SummerFun20 | 6.6.2020 | 12 days | CM100 | Roberta | 1 | TV | Local | 50% |
| 111 | SummerFun20 | 6.6.2020 | 12 days | CM100 | Roberta | 2 | TV | National | 50% |
| 222 | SummerZing20 | 6.8.2020 | 30 days | CM101 | Sue | 1 | TV | Local | 60% |
| 222 | SummerZing20 | 6.8.2020 | 30 days | CM101 | Sue | 3 | Radio | Local | 30% |
| 222 | SummerZing20 | 6.8.2020 | 30 days | CM101 | Sue | 5 | Print | Local | 10% |
| 333 | FallBall20 | 6.9.2020 | 12 days | CM102 | John | 3 | Radio | Local | 80% |
| 333 | FallBall20 | 6.9.2020 | 12 days | CM102 | John | 4 | Radio | National | 20% |
| 444 | AutmnStyle20 | 6.9.2020 | 5 days | CM103 | Nancy | 6 | Print | National | 100% |
| 555 | AutmnColors20 | 6.9.2020 | 3 days | CM100 | Roberta | 3 | Radio | Local | 100% |

## Given FDs:

**A→B**
**A→C**
**A→D**
**A→E**
**A→F**
**E→F**
**G→H**
**G→I**
**{A,G}→J**

## Solution:

- First Find closure of $\{ABCDEFGHIJ\}^+$=?
- $\{ABCDEFGHIJ\}^+ = \{ABCDEFGHIJ\}$
- As A→B,C,D,E,F, so remove them from left side. We left with {AGHIJ}
- Now $\{AGHIJ\}^+ = \{ABCDEFGHIJ\}$



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# Normalise Relation (AD CAMPAIGN MIX) into 2NF (2)

- As G→H,I, So remove **H** and **I** from left side. We left with {AGJ}
- Now $\{AGJ\}^+ = \{ABCDEFGHIJ\}$
- As {A,G}→J, So remove J from left side. We left with {AG}
- Now $\{AG\}^+ = \{ABCDEFGHIJ\}$
- As {AG} determines all attributes of the relation, therefore, AG is superkey.
- Moreover, no proper subset of {AG} can determine all attributes. Therefore, {AG} is also candidate key.
- Prime attributes = {A,G}
- Non-prime attributes = {B,C,D,E,F,H,I,J}

# Normalise Relation (AD CAMPAIGN MIX) into 2NF (3)

**A** is prime attribute and determining non-prime attributes, therefore, these are PFD

**A→B**
**A→C**
**A→D**
**A→E**
**A→F**

**G** is prime attribute and determining non-prime attributes, therefore, these are PFD

**G→H**
**G→I**

# Normalise Relation (AD CAMPAIGN MIX) into 2NF (4)

- Find $\{A\}^+=?$
- $\{A\}^+=\{ABCDEF\}$
- Assign these attributes to a sub-relation R1(A,B,C,D,E,F)
- Find $\{G\}^+=?$
- $\{G\}^+=\{GHI\}$
- Assign these attributes to a sub-relation R2(G,H,I)
- Now assign the remaining attribute of the relation to a to a sub-relation R3(J)

# Normalise Relation (AD CAMPAIGN MIX) into 2NF (5)

- R(A,B,C,D,E,F,G,H,I,J)

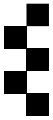
is decomposed

- R1(A,B,C,D,E,F)
- R2(G,H,I)
- R3(J)

# Lossless Join Decomposition

# And

# Dependency Preserving Decomposition



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# Lossless Join Decomposition

- Decomposition of a relation into sub-relations such that a natural join of the sub-relations yields back the original relation.
- If we decompose a relation R into relations R1 and R2,
  - Decomposition is **lossy** if $R1 \bowtie R2 \supset R$
  - Decomposition is **lossy** if $R1 \bowtie R2 \subset R$
  - Decomposition is **lossless** if $R1 \bowtie R2 = R$

# Conditions for Lossless Join Decomposition

- To check for lossless join decomposition using given set of FD, following conditions must hold:

  - **Union of Attributes** of R1 and R2 must be equal to attribute of R.
  - **Intersection of Attributes** of R1 and R2 must not be NULL.
  - **Common attribute** must be a key for at least one relation (R1 or R2)

# Dependency Preserving Decomposition

- If we decompose a relation R into relations R1 and R2, All dependencies of R either must be a part of R1 or R2 or must be derivable from combination of FD's of R1 and R2.
- For Example,

  A relation R (A, B, C, D) with FD set{A->BC}

  is decomposed into

  *R1(ABC) and R2(AD)*

  which is dependency preserving because FD A->BC is a part of R1(ABC).

# Normalise Relation (AD CAMPAIGN MIX) into 2NF (6)

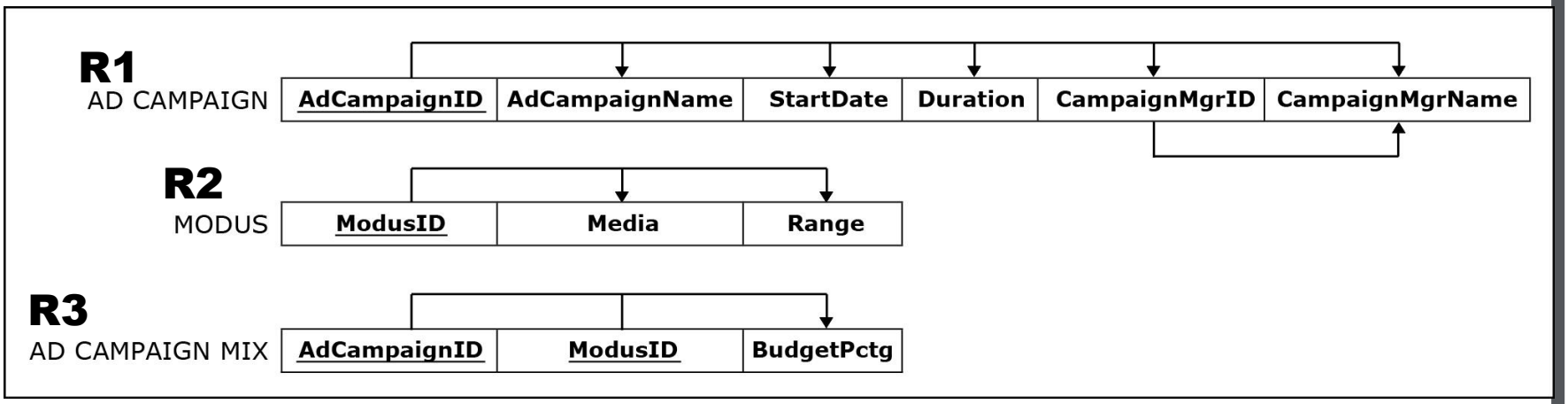
- Now, join sub-relations R1(A,B,C,D,E,F), R2(G,H,I), R3(J)
- Join should be lossless, therefore, there should be a common attribute which must be a candidate of any sub-relation R1 or R3. As {A} is a candidate key of R1, We will get
- R1(A,B,C,D,E,F) and R3(A,J)
- Similarly, {G} is candidate key of R2, so we will get
- R2(G,H,I) and R3(A,G,J)
- So, the lossless join of R(A,B,C,D,E,F,G,H,I,J) IS

  R1(A,B,C,D,E,F)  R2(G,H,I)  R3(A,G,J)

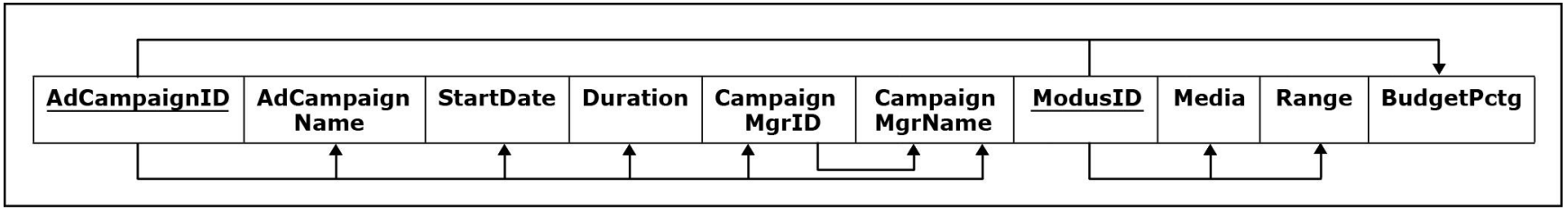
- Now relation R is in 2NF.

# Normalise Relation (AD CAMPAIGN MIX) into 2NF (6)

**R1 is AD COMPAIGN**
**R2 is MODUS**
**R3 is AD CAMPAIN MIX**




*Original Relation **R** (for the reference)*




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